

"Math Is Cool" Championships — 2025-26

7th/8th grade — January/February 2026

Sponsored by:

GENERAL INSTRUCTIONS applying to all tests:

- *Good sportsmanship is expected throughout the competition by all involved (competitors and observers). Display of poor sportsmanship will result in disqualification.*
- *Competitors may not use calculators or any other aids on any portion of this contest.*
- *Unless stated otherwise:*
 - *All answers are integers, and any non-integer answers will be "coded" as integers.*
 - *All fractions and ratios must be reduced to simplest form, all radicals must be simplified, and all denominators must be rationalized.*
 - *Do not round or approximate answers. Leave answers in terms of π or other irrational quantities (e.g., $\sqrt{2}$), where applicable.*
- *Units are not necessary as part of your answer, However, if you choose to use units, they must be correct.*
- *Record all answers on the colored cover sheets in the answer column only.*
- ***Be sure that the student name, school, team number, etc. has been filled out at the top of each answer sheet.***
- *Tests will be scored as a 0 if answers are not recorded correctly on the answer sheets.*
- *Blank answer sheets and answer sheets with no name will be scored as a 0.*

FINAL SCORES AND AWARDS

Individual awards are determined by both the Mental Math and Individual Test scores. Individual ties are broken based on the following, in this order: total scaled individual points, total number of correct answers on the Individual Test, Mental Math raw score, number of correct answers from Individual Test #31-40, number of correct answers from Individual Test #16-30, highest numbered question answered correctly on the Individual Test working backwards from #40.

Team (School) awards are based on the highest score from amongst each of the school's "teams of 4 students" in each event and is calculated as $2 \cdot (\text{Sum of highest 3 Mental Math scores}) + 2 \cdot (\text{Multiple Choice}) + 6 \cdot (\text{Team}) + 1 \cdot (\text{Triple Jump}) + 1 \cdot (\text{College Bowl})$, for approximate weights of 25%, 20%, 30%, 15% and 10% respectively. Team ties are broken based on highest event score in order of the events, starting with Mental Math.

MENTAL MATH TEST - 30 sec./quest., 8 problems, ~8%/25% of individ./team scores

The proctor will read each question twice. You may not do any writing or talking while arriving at a solution. Record only your answer on your answer sheet. You may not change, cross out, erase, or write over an answer once you have written it down. The maximum wait time is 30 seconds after completion of the second reading of the question. Correct answers receive 1 point.

INDIVIDUAL TEST - 35 minutes, 40 problems, ~92% of individual score

When you are prompted to begin, tear off the colored answer sheet and begin testing. No talking during this individual test. You will be given a 5 minute time warning. Correct answers receive 2 points for problems 1-30 and 3 points for 31-40 (in the scaled score).

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7th/8th Grade — January/February 2026

Final Score (out of 8)

Room #

School Name

Student Name

Team #

Mental Math - ~25% of team score & ~8% of individual score

All students in the room will concurrently be asked the same eight questions in this individual test. When it is time to begin, the proctor will read the first question twice. You may not do any writing or talking while arriving at a solution. Once you have a solution, record it on the sheet in front of you. **You may not change or cross out answers once you have written an answer down. If there are eraser marks, write-overs, or crossed-out answers, they will be marked wrong.** Once all students have laid their pencils on the desk, another question will be asked. If a student doesn't lay his or her pencil down, the maximum wait time is 30 seconds after completion of the second reading of the question before the next question is read. You may continue to work on a problem (in your head) while the next question is being read. The raw score is 1 point per correct answer.

STUDENT: DO NOT WRITE IN SHADED REGIONS (or anywhere else, other than the answer box)

Answer		Scorer 2	Scorer 1
		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
7 th /8 th Grade	TOTAL:		

"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Key

Mental Math Contest – Answer Key

30 seconds per question - ~25% of team score & ~8% of individual score

SCORERS — Write-overs, Cross-outs, and Erasures Must be Marked Incorrect (0)
Bracketed items [...] in the answer key are optional.

7th/8th Grade

Answer		
1	50 [= mean]	What is the mean of the numbers forty, sixty, eighty, and twenty?
2	350 [millimeters]	How many millimeters are in zero-point-three-five meters?
3	21 [candies]	A bag contains only pink and purple candies. All but twelve of the candies are pink and all but nine of the candies are purple. How many candies are in the bag?
4	34 [inches]	In inches, what is the perimeter of a rectangle whose length is ten-point-five inches and whose width is six-point-five inches?
5	4 [ways]	How many ways are there to arrange the letters in the word SASS, spelled S-A-S-S?
6	480 [seconds]	Anderson completes a one-mile race in six minutes. In this race, Brody runs at an average rate that is three-fourths as fast as Anderson. In seconds, how long does it take Brody to complete the one-mile race?
7	1 [= 6 th term]	The rule in a certain number sequence is: divide the first term by one to get the second term, divide the second term by two to get the third term, divide the third term by three to get the fourth term, and so on. If the first term in this sequence is one hundred twenty, what is the sixth term?
8	1080 [= LCM]	What is the least common multiple of ten, twenty-four, and fifty-four?

"Math Is Cool" Championships — 2025-26

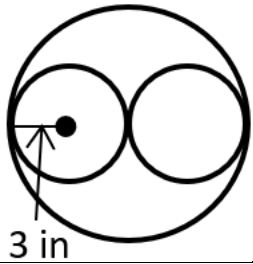
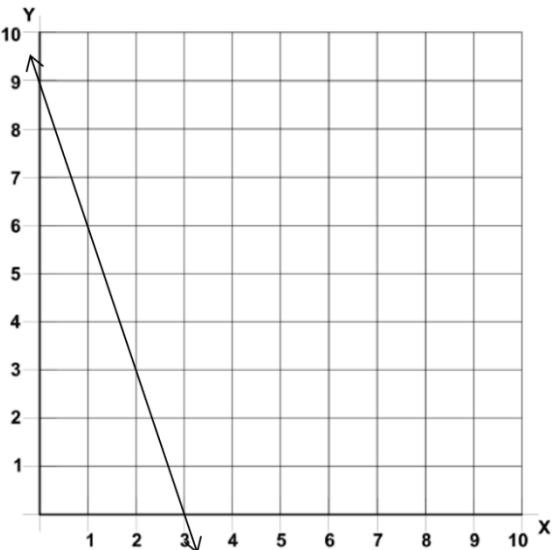
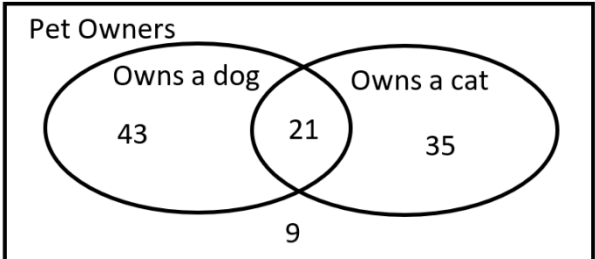
7th/8th Grade — January/February 2026

Individual Contest

Record all answers on the colored cover sheet. 35 minutes, 40 problems, ~92% of individual score.

No talking during this individual test. A 5-minute time warning will be given.

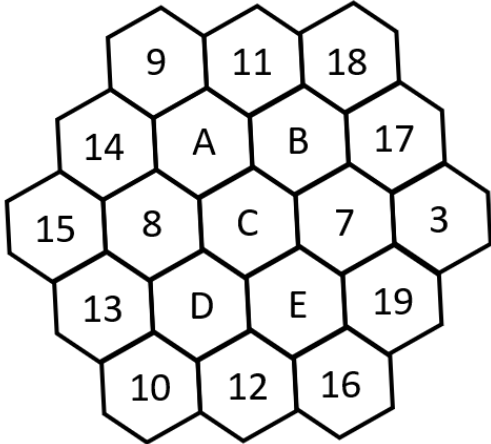
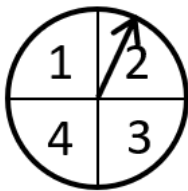
	Questions 1-30: 2 points each
1	Evaluate: $901 - 82 + 3$
2	What is the number of square inches in the area of a rectangle that has a width of 13 inches and a length of 31 inches?
3	Billie walks at an average rate of 3 miles per hour. How many minutes does it take her to walk 2 miles?
4	A jar has 35 green marbles and 30 blue marbles. As a simplified fraction, the probability that a randomly selected marble is green is A/B . What is $A + B$?
5	Solve for x : $29x + 53 = 256$
6	What is the median of the data set? {31, 28, 33, 35, 26, 39, 30}
7	Ninety percent of the students at Kenroy Elementary School are right-handed. If there are 170 students at Kenroy Elementary, how many are right-handed?
8	How many inches are equal to 1.75 feet?
9	A negative integer, P , is 19 units away from a positive integer, Q , on the number line. What is the largest possible value of Q ?
10	Avril has seven different pairs of pants, six different shirts, and four different pairs of shoes. How many different outfits are available to Avril consisting of one pair of pants, one shirt, and one pair of shoes?
11	The first three terms in an arithmetic sequence are -2, 7, and 16. What is the fifth term in the sequence?
12	What is the sum of the prime numbers between 24 and 32?
13	Evaluate: $12 \cdot 3^4$
14	A bag contains blue, green, and purple candies. Three-eighths of the candies are blue, $1/4$ of the candies are green, and there are 24 purple candies. How many total candies are in the bag?
15	The letters A, C, D, and R, are arranged randomly. As a simplified fraction, the probability that they are arranged in the order of the word CARD is P/Q . What is $P + Q$?
Continued on next page.	

16	Two congruent circles are externally tangent to each other, and both are internally tangent to a third larger circle as shown. The centers of all three circles lie on a diameter of the larger circle. The radius of one of the congruent circles is 3 inches. In terms of π , the area of the large circle is $A\pi$ in ² . What is A?	
17	Using only quarters (25¢), dimes (10¢), nickels (5¢), and/or pennies (1¢), what is the fewest number of coins needed to make 93¢?	
18	There are 2 points on the line shown that are in the first quadrant and that have integer coordinates (x, y). What is the sum of the y-coordinates of these 2 points?	
19	The first three terms of a geometric sequence are 5, 15, and 45. What is the fourth term?	
20	A palindrome is a number that reads the same forwards as backwards, like 101. What is the largest 5-digit palindrome in which the ones, tens, and hundreds digits are all different from each other?	
21	How many ways are there to arrange the letters in the word JUNIPER, if the first letter is either E, I, or U, and the remaining letters are in alphabetical order?	
22	According to the Venn Diagram shown and as a simplified fraction, the probability that a dog-owner also owns a cat is A/B . What is $A + B$?	
23	Animal can wash 24 plates in 10 minutes and Gonzo can wash 18 plates in 15 minutes. Working together, how many plates can Animal and Gonzo wash in 20 minutes?	
24	An infinite sequence follows the rule that the 2 nd term equals the 1 st term times 1, the 3 rd term equals the 2 nd term times 3, the 4 th term equals the 3 rd term times 5, and so on multiplying each successive term by the next consecutive odd integer to get the following term. If the first term is 4, what is the 6 th term?	
25	Let $a \nmid b = \frac{(a^2+b)}{(b^2-a)}$. Then, as a simplified fraction $12 \nmid 6 = A/B$. What is $A + B$?	

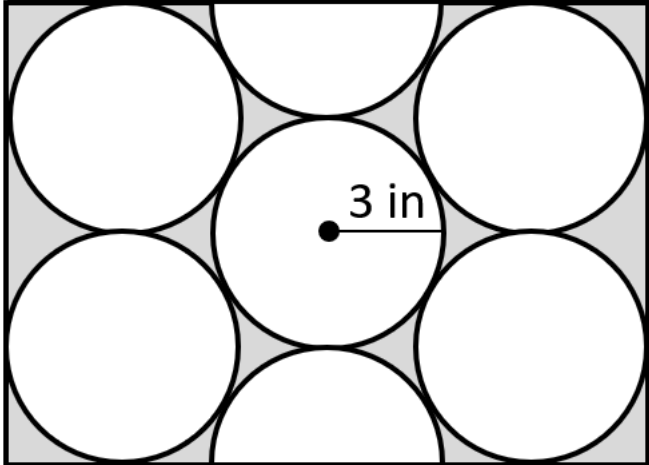
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26	The vertices of a triangle are each labeled with a distinct odd, single-digit positive integer. The sides of the triangle are each labeled with the product of the integers at their endpoints. What is the largest possible sum of the labels of the three sides?
27	Emery eats $\frac{1}{4}$ and Liam eats $\frac{1}{6}$ of a cake. Then Vienna eats $\frac{1}{7}$ of what remains. Then Zeke eats $\frac{2}{3}$ of what remains after Vienna eats her portion. Then Ariel eats the rest. As a simplified fraction, Ariel eats A/B of the cake. What is $A + B$?
28	A set of seven distinct positive integers has a mean of 12, which is also the median. What is the largest possible range, R , where R = the largest integer minus the smallest integer in the set.
29	A rectangle has integer side lengths and an integer diagonal length. If the area of the rectangle is 120 cm^2 , what is the number of centimeters in the length of the diagonal?
30	The expression $\frac{(a^5b^7c^9)^2(a^3b^4c^7)^4}{ab^2c^4}$ can be written as $(a^x b^y c^z)^7$. What is the value of $x + y + z$?

Challenge Questions: 3 points each

31	The five-digit integer 49A70 is divisible by 30, where A represents a non-negative single-digit integer. What is the sum of all possible values of A?
32	In the Magic Hexagon shown, the consecutive integers 1 through 19 are distributed such that the sum of all 5 horizontal rows, and of both sets of 5 diagonal rows (the ones sloping up to the right and the ones sloping down to the right) is 38. There is exactly one set of 5 distinct integers that can replace the letters A, B, C, D, and E to ensure the sum of 38 in all rows and diagonals. These integers when listed in order ABCDE create a 5-digit integer. What is this 5-digit integer? 
33	The digits 3 and 5 are used to create 8 distinct three-digit positive integers. These three-digit integers can consist of all 3s, all 5s, or a combination of 3s and 5s. What is the mean of these 8 integers?
34	The spinner shown, which is divided into 4 equal sections, is spun 5 times. As a simplified fraction, the probability that all four numbers are spun at least once is A/B. What is $A + B$? 
35	Alek and Aurelia are walking due east side-by-side at the same rate. Alek steps onto an eastward moving walkway and continues walking due east at the same rate as before, while Aurelia increases her rate by 24%. When Alek reaches the end of the moving walkway, Aurelia is 42% of the way to the end of the walkway. As a simplified fraction, the ratio of Alek and Aurelia's original walking rate to the rate of the walkway is A/B . What is $A + B$?

Continued on next page.

36	The graph of $y = -x + 2$ intersects with the graph of the parabola $y = \frac{1}{2}x^2 - 2$ in two points (x, y) . The distance between these two points is $A\sqrt{B}$ units, where A and B are single-digit integers and B has no perfect square factors other than 1. What is $A + B$?
37	How many ordered pairs of positive integers satisfy the following equation? $\frac{5}{a} - \frac{6}{b} = \frac{1}{2}$
38	In a geometric sequence of positive numbers, if the 21 st term is 500 and the 39 th term is 32, what is the 33 rd term?
39	In the figure shown below there are five complete circles and two partial circles all with the same radius that are externally tangent to each other and internally tangent to a rectangle. Expressed in terms of π, the area of the shaded region is $A\sqrt{3} + B - C\pi$ in², where A, B and C are positive integers. What is $A + B + C$? 
40	Let N be an integer whose prime factorization is in the form $a^3 \cdot b^2 \cdot c$, where a , b , and c are distinct positive prime numbers. How many values of N are less than 900?

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KEY

Individual Contest - Answer Key

SCORERS: Bracketed [...] items in answer key are optional. Just mark the score as 0 or 1 and add up those values to reflect total correct.
First Scorer - use the right-hand columns so 2nd scorer can do a blind scoring.

	Answer
1	822
2	403 [in ²]
3	40 [minutes]
4	[A + B =] 20
5	[x =] 7
6	31 [= median]
7	153 [students]
8	21 [inches]
9	[Q =] 18
10	168 [outfits]
11	34 [= 5 th term]
12	60
13	972
14	64 [candies]
15	[P + Q =] 25

	Answer
16	[A =] 36
17	8 [coins]
18	9
19	135
20	98789
21	3 [ways]
22	[A + B =] 85
23	72 [plates]
24	3780 [= 6 th term]
25	[A + B =] 29
26	143 [= sum]
27	[A + B =] 7
28	[R =] 38
29	17 [cm]
30	[x + y + z =] 13

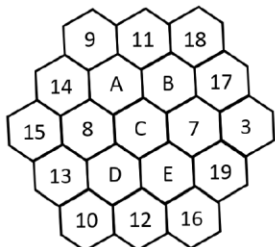
	Answer
31	12 [= sum]
32	61542 [= ABCDE]
33	444 [= mean]
34	[A + B =] 79
35	[A + B =] 62
36	[A + B =] 8
37	7 [ordered pairs]
38	80 [= 33 rd term]
39	[A + B + C =] 198
40	6 [values]

7th/8th Grade
January/February 2026

- 1 Evaluate: $901 - 82 + 3$
- 2 What is the number of square inches in the area of a rectangle that has a width of 13 inches and a length of 31 inches?
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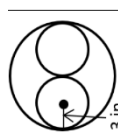


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- 35 Alek and Aurelia are walking due east side-by-side at the same rate. Alek steps onto an eastward moving walkway and continues walking due east at the same rate as before, while Aurelia increases her rate by 24%. When Alek reaches the end of the moving walkway, Aurelia is 42% of the way to the end of the walkway. As a simplified fraction, the ratio of Alek and Aurelia's original walking rate to the rate of the walkway is A/B . What is $A + B$?

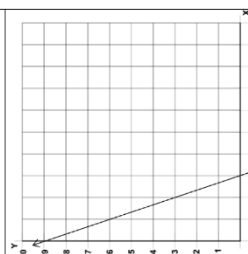


Two congruent circles are externally tangent to each other, and both are internally tangent to a third larger circle as shown. The centers of all three circles lie on a diameter of the larger circle. The radius of one of the congruent circles is 3 inches. In terms of π , the area of the large circle is $A\pi$ in². What is A ?

16

Using only quarters (25¢), dimes (10¢), nickels (5¢), and/or pennies (1¢), what is the fewest number of coins needed to make 93¢?

There are 2 points on the line shown that are in the first quadrant and that have integer coordinates (x, y) . What is the sum of the y -coordinates of these 2 points?

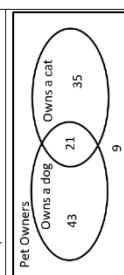


The first three terms of a geometric sequence are 5, 15, and 45. What is the fourth term?

A palindrome is a number that reads the same forwards as backwards, like 101. What is the largest 5-digit palindrome in which the ones, tens, and hundreds digits are all different from each other?

How many ways are there to arrange the letters in the word JUNDPER, if the first letter is either E, I, or U, and the remaining letters are in alphabetical order?

According to the Venn Diagram shown and as a simplified fraction, the probability that a dog-owner also owns a cat is A/B . What is $A + B$?



Animal can wash 24 plates in 10 minutes and Gonzo can wash 18 plates in 15 minutes. Working together, how many plates can Animal and Gonzo wash in 20 minutes?

An infinite sequence follows the rule that the 2nd term equals the 1st term times 1, the 3rd term equals the 2nd term times 3, the 4th term equals the 3rd term times 5, and so on multiplying each successive term by the next consecutive odd integer to get the following term. If the first term is 4, what is the 6th term?

Let $a \vee b = \frac{(a^2+b^2)}{(b^2-a)}$. Then, as a simplified fraction $12 \vee 6 = A/B$. What is $A + B$?

The vertices of a triangle are each labeled with a distinct odd, single-digit positive integer. The sides of the triangle are each labeled with the product of the integers at their endpoints. What is the largest possible sum of the labels of the three sides?

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The expression $\frac{(a^2b^7c^9)^2(a^3b^4c^7)^4}{ab^2c^4}$ can be written as $(a^x b^y c^z)^7$. What is the value of $x + y + z$?

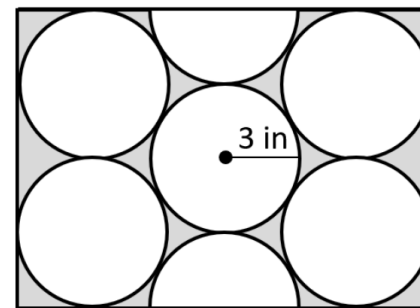
- 36 The graph of $y = -x + 2$ intersects with the graph of the parabola $y = \frac{1}{2}x^2 - 2$ in two points (x, y) . The distance between these two points is $A\sqrt{B}$ units, where A and B are single-digit integers and B has no perfect square factors other than 1. What is $A + B$?

- 37 How many ordered pairs of positive integers satisfy the following equation?

$$\frac{5}{a} - \frac{6}{b} = \frac{1}{2}$$

- 38 In a geometric sequence of positive numbers, if the 21st term is 500 and the 39th term is 32, what is the 33rd term?

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- 40 Let N be an integer whose prime factorization is in the form $a^3 \cdot b^2 \cdot c$, where a , b , and c are distinct positive prime numbers. How many values of N are less than 900?

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Total Correct (all columns)

Room #

SCHOOL NAME

STUDENT NAME

Team #

Individual Contest - Score Sheet

STUDENTS: DO NOT WRITE IN SHADED REGIONS

	Answer	1 or 0	1 or 0
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
1-15 TOTAL:			

	Answer	1 or 0	1 or 0
16			
17			
18			
19			
20			
21			
22			
23			
24			
25			
26			
27			
28			
29			
30			
16-30 TOTAL:			

	Answer	1 or 0	1 or 0
31			
32			
33			
34			
35			
36			
37			
38			
39			
40			
31-40 TOTAL:			

7th/8th Grade

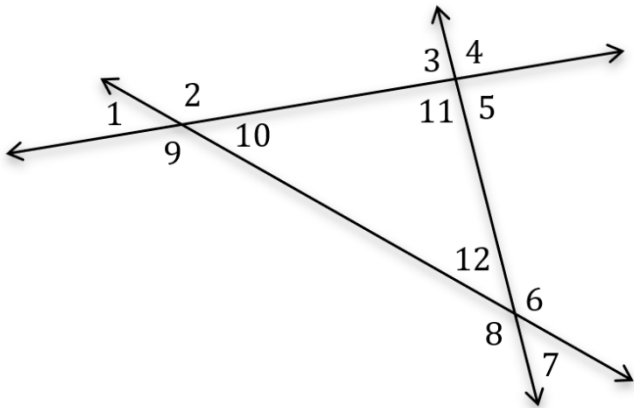
January/February 2026

Scorers: Just score as 0 or 1 and add up those values (i.e., just work with number correct).

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7th/8th Grade — January/February 2026

Team Multiple Choice Contest

1	Evaluate the expression: $2^1 + 2^2 + 2^3 + 2^4 + 2^5$ A) 26 B) 30 C) 61 D) 62 E) None of the above
2	Evaluate the expression: $10^1 - 9^2 + 8^3 - 7^4 + 6^5$ A) 18 B) 1394 C) 4816 D) 6874 E) None of the above
3	Evaluate the expression: $25^2 + 15^4 + 5^6 - 23^2 - 13^4 - 3^6$ A) 84 B) 37056 C) 38336 D) 39616 E) None of the above
	REFER TO THE FIGURE SHOWN HERE, CONSISTING OF THREE INTERSECTING LINES, TO SOLVE PROBLEMS #4 THROUGH #6. 
4	If $m\angle 1 = 38^\circ$ and $m\angle 4 = 98^\circ$, what is $m\angle 12$? A) 34° B) 36° C) 44° D) 54° E) None of the above
5	If $m\angle 2 = (17n + 3)^\circ$, $m\angle 3 = (11n - 5)^\circ$, and $m\angle 7 = (4n + 10)^\circ$, then what is $m\angle 6$? Note: this may result in a different set of angle measures from the ones in problem 4. A) 42° B) 98° C) 128° D) 136° E) 138°
6	If $m\angle 11 > m\angle 12 > m\angle 10$, then the triangle formed by the three intersecting lines in the figure is a triangle in which all three sides are different lengths, i.e. a scalene triangle. If the side that is neither shortest nor longest is 18 cm in length, then how many distinct sets of integer side lengths, measured in centimeters, are possible for this triangle? A) 120 B) 136 C) 144 D) 153 E) infinite
Continued on next page.	

USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #7 THROUGH #10.

An arithmetic sequence is a sequence in which each term after the first is obtained by adding a constant to the previous term. For example, 3, 9, 15, 21, 27 is an arithmetic sequence with five terms, in which the first term is 3 and the constant added is 6, also known as the common difference.

A geometric sequence is a sequence in which each term after the first is obtained by multiplying the previous term by a constant. For example, 6, 12, 24, 48, 96 is a geometric sequence with five terms, in which the first term is 6 and the constant that terms are multiplied by is 2, also known as the common ratio.

7 The first term of an arithmetic sequence is 23. What is the common difference that results in a fifth term of 41?

- A) 4.5 B) 5 C) 5.5 D) 6 E) None of the above

8 The common ratio of a geometric sequence is $\frac{2}{3}$. What is the first term in the sequence if the second term is 10?

- A) $\frac{20}{3}$ B) 15 C) 18 D) 20 E) None of the above

9 The first row in this 4×4 array is an arithmetic sequence with 4 terms, and each column is a geometric sequence with 4 terms. Three of the squares are labeled with A, B, and C, as shown. What is $A + B + C$?

- A) 93 B) 128 C) 137 D) 143 E) 154

1	A		7
5		B	14
		45	
C	192		

10 Consider the sequence of positive integers

1, 2, 1, 3, 6, 3, 1, 2, 5, 8, 5, 2, 1, 3, 6, 9, 12, 9, 6, 3, 1, 2, 5, 8, 11, 14, 11, 8, 5, 2, 1, ...

The rule for the sequence is that between the first pair of 1s is a 2, then between the next pair of 1s is 363, then between the next pair of 1s is 25852, and so on. The set of numbers between each successive pair of 1s alternates between an arithmetic sequence starting with 2 and an arithmetic sequence starting with 3, both of which have a common difference of 3 until reaching the middle number in the set, followed by a common difference of -3.

What is the 2026th term in this sequence?

- A) 1 B) 132 C) 133 D) 134 E) 135

"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Key

Team Multiple Choice Contest – Answer Key

7th/8th Grade

Correct responses are worth 2 points, incorrect responses are worth -1 point, and absence of a response is worth 0 points.

Answer	
1	D
2	E [5816]
3	B
4	C
5	E
6	B
7	A
8	B
9	D
10	D

USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #7 THROUGH #10.

An arithmetic sequence is a sequence in which each term after the first is obtained by adding a constant to the previous term. For example, 3, 9, 15, 21, 27 is an arithmetic sequence with five terms, in which the first term is 3 and the constant added is 6, also known as the common difference.

A geometric sequence is a sequence in which each term after the first is obtained by multiplying the previous term by a constant. For example, 6, 12, 24, 48, 96 is a geometric sequence with five terms, in which the first term is 6 and the constant that terms are multiplied by is 2, also known as the common ratio.

The first term of an arithmetic sequence is 23. What is the common difference that results in a fifth term of 41?

- A) 4.5 B) 5 C) 5.5 D) 6 E) None of the above

The common ratio of a geometric sequence is $\frac{2}{3}$. What is the first term in the sequence if the second term is 10?

- A) $\frac{20}{3}$ B) 15 C) 18 D) 20 E) None of the above

The first row in this 4×4 array is an arithmetic sequence with 4 terms, and each column is a geometric sequence with 4 terms. Three of the squares are labeled with A, B, and C, as shown. What is $A + B + C$?

1	A		7
5		B	14
			45
C	192		

- A) 93 B) 128 C) 137 D) 143 E) 154

Consider the sequence of positive integers

1, 2, 1, 3, 6, 3, 1, 2, 5, 8, 5, 2, 1, 3, 6, 9, 12, 9, 6, 3, 1, 2, 5, 8, 11, 14, 11, 8, 5, 2, 1, ...

The rule for the sequence is that between the first pair of 1s is a 2, then between the next pair of 1s is 363, then between the next pair of 1s is 25852, and so on. The set of numbers between each successive pair of 1s alternates between an arithmetic sequence starting with 2 and an arithmetic sequence starting with 3, both of which have a common difference of 3 until reaching the middle number in the set, followed by a common difference of -3.

What is the 2026th term in this sequence?

- A) 1 B) 132 C) 133 D) 134 E) 135

1

Evaluate the expression: $2^1 + 2^2 + 2^3 + 2^4 + 2^5$

- A) 26 B) 30 C) 61 D) 62 E) None of the above

2

Evaluate the expression: $10^1 - 9^2 + 8^3 - 7^4 + 6^5$

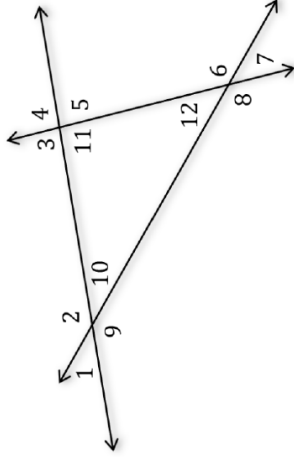
- A) 18 B) 1394 C) 4816 D) 6874 E) None of the above

3

Evaluate the expression: $25^2 + 15^4 + 5^6 - 23^2 - 13^4 - 3^6$

- A) 84 B) 37056 C) 38336 D) 39616 E) None of the above

REFER TO THE FIGURE SHOWN HERE, CONSISTING OF THREE INTERSECTING LINES, TO SOLVE PROBLEMS #4 THROUGH #6.



4 If $m\angle 1 = 38^\circ$ and $m\angle 4 = 98^\circ$, what is $m\angle 12$?

- A) 34° B) 36° C) 44° D) 54° E) None of the above

5

If $m\angle 2 = (17n + 3)^\circ$, $m\angle 3 = (11n - 5)^\circ$, and $m\angle 7 = (4n + 10)^\circ$, then what is $m\angle 6$? Note: this may result in a different set of angle measures from the ones in problem 4.

- A) 42° B) 98° C) 128° D) 136° E) 138°

6

If $m\angle 11 > m\angle 12 > m\angle 10$, then the triangle formed by the three intersecting lines in the figure is a triangle in which all three sides are different lengths, i.e. a scalene triangle. If the side that is neither shortest nor longest is 18 cm in length, then how many distinct sets of integer side lengths, measured in centimeters, are possible for this triangle?

- A) 120 B) 136 C) 144 D) 153 E) infinite

"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Final Score (out of 20)

Room #

School Name

Team #

Team Multiple Choice Contest - 15 minutes - ~20% of team score

*This test is the only test where you will be penalized for incorrect responses. You will receive two points for a correct letter response, zero points for leaving it blank, and minus one point for an incorrect response. When you are prompted to begin, tear off the colored answer sheet, pass out a copy of the test to each team member, and begin testing. **ONLY a letter response should be listed as an answer on this answer sheet.***

Correct responses are worth 2 points, incorrect responses are worth -1 point, and absence of a response is worth 0 points.

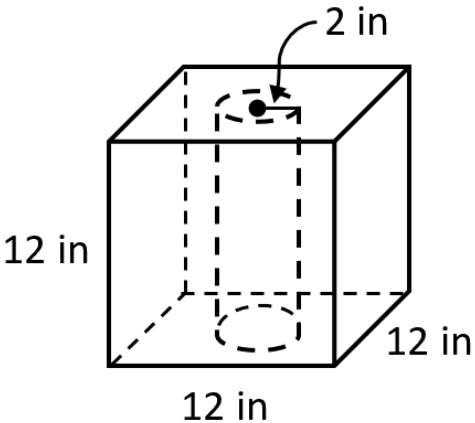
STUDENTS: DO NOT WRITE IN SHADED REGIONS

Answer		Scorer 2	Scorer 1
		-1, 0, or 2	-1, 0, or 2
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
7 th /8 th Grade		TOTAL:	

"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Team Contest

1	A plastic jug contains 4 red, 6 blue, and 8 purple balls. As a simplified fraction, the probability that a randomly selected ball is not purple is A/B . What is $A + B$?
2	Evaluate: $2^4 - 6 \cdot \sqrt{25} + \frac{90}{6}$
3	Bert painted one-third of a fence, and Ernie painted one-fourth of the same fence, each at a constant rate. If Bert, continuing at the same rate, finished painting the rest of the fence in 20 minutes, how many minutes would it have taken him to paint the entire fence by himself?
4	How many positive 2-digit multiples of 7 are there?
5	A data set consists of 8 distinct positive odd integers and has a range of 20. One of the integers is 15. What is the largest possible median for this data set?
6	Solve for x : $11 - 4x = x + 125 - 3x$
7	A cube has a side length of 12 inches. A cylindrical hole with a radius of 2 inches is drilled through the center of the top and the bottom of the cube. In square inches, the surface area of the solid, including the surface of the cylindrical hole, is $A + B\pi$, where A and B are positive integers. What is $A + B$? 
8	Consider the sequence of numbers 1, 6, 16, 36, 76, $\dots$, where each number after 1 is four more than two times the previous number. The positive difference between the 31st and 32nd numbers in the sequence can be written as $5 \cdot p^a$, where p is a prime number and a is an integer. What is the value of $p + a$?
9	One morning each member of Gregorio's team at work drank a 12-ounce mixture of coffee with milk. The amounts of coffee and milk varied from mug to mug but were never zero. Gregorio drank one-ninth of the total amount of milk and one-eleventh of the total amount of coffee. How many people are on his team?
10	Each of the 24 students in a math class is assigned a number from 1 to 24 corresponding to a numbered pocket in a classroom phone caddy with 24 pockets. The caddy accidentally falls to the floor and all the phones fall out. The classroom assistant replaces all of the phones to best of his ability, but some of the phones may be in the wrong pocket. In how many ways can exactly 3 of the phones be in the wrong pocket?

"Math Is Cool" Championships — 2025-26

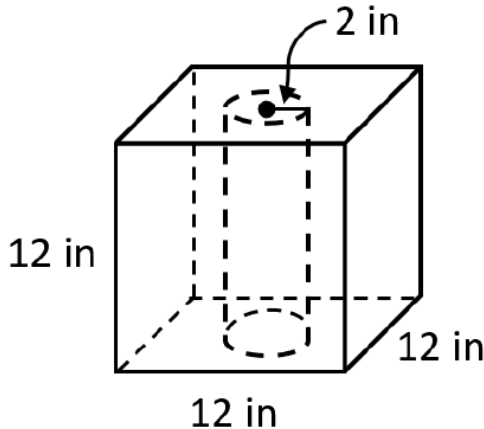
7th/8th Grade — January/February 2026

Key

Team Contest - Answer Key

7th/8th Grade

Answer	
1	$[A + B =] 14$
2	1
3	48 [minutes]
4	13 [multiples]
5	28
6	$[x =] -57$
7	$[A + B =] 904$
8	$[p + a =] 32$
9	10 [people]
10	4048 [ways]

- | | |
|----|--|
| 1 | A plastic jug contains 4 red, 6 blue, and 8 purple balls. As a simplified fraction, the probability that a randomly selected ball is not purple is A/B . What is $A + B$? |
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"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Final Score (out of 10)

Room #

School Name

Team #

Team Contest - 15 minutes - ~30% of team score

When you are prompted to begin, tear off the colored answer sheet and give a copy of the test to each of your team members and begin testing. Each problem is scored as a 1 or 0. Record all answers on this colored answer sheet.

STUDENTS: DO NOT WRITE IN SHADED REGIONS

Answer		Scorer 2	Scorer 1
		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
7 th /8 th Grade		TOTAL:	

"Math Is Cool" Championships — 2025-26

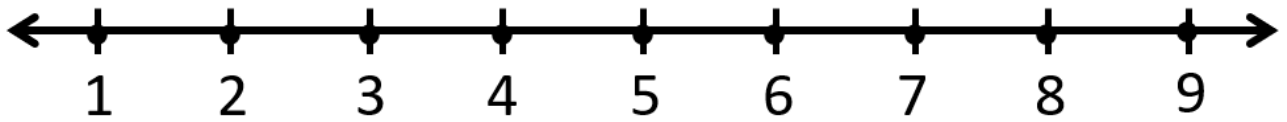
7th/8th Grade — January/February 2026

Linda Moore Triple Jump

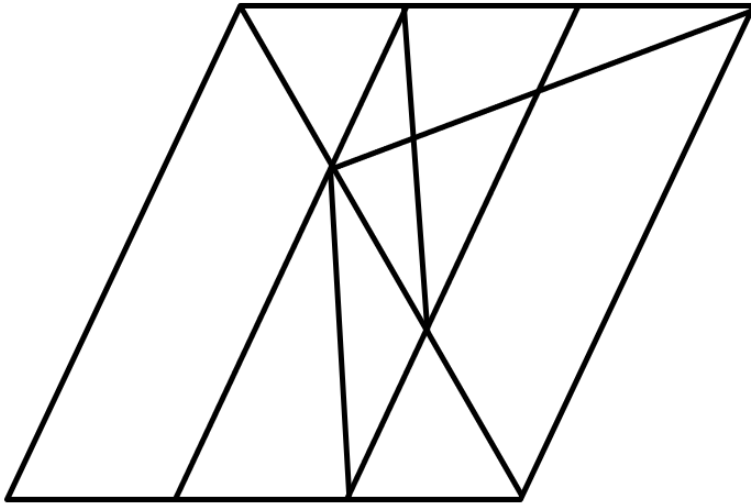
1	What is 80% of 22.5?																
2	The integers 1 through 10 are written as English words and listed in order alphabetically. As an integer, what is the ninth word in the list?																
3	It takes 3 workers 20 minutes to wash 80 plates. Working at the same rate, how many plates can 5 workers wash in 15 minutes?																
4	A cylinder has a radius of 15 millimeters and a height of 0.12 meters. In terms of π and in cubic <i>centimeters</i> , the volume of the prism can be expressed as $A\pi \text{ cm}^3$. What is the value of A?																
5	In a survey, people were asked about their favorite candy. The number of people who indicated that Skittles is their favorite candy is shown by age group in the histogram below. What is the maximum possible median age in an integer number of years for those who prefer Skittles from the survey? Favorite Candy is Skittles <table border="1"> <thead> <tr> <th>Age Group in years</th> <th>Number of People</th> </tr> </thead> <tbody> <tr> <td>5 - 9</td> <td>1</td> </tr> <tr> <td>10 - 14</td> <td>2</td> </tr> <tr> <td>15 - 19</td> <td>4</td> </tr> <tr> <td>20 - 24</td> <td>7</td> </tr> <tr> <td>25 - 29</td> <td>5</td> </tr> <tr> <td>30 - 34</td> <td>3</td> </tr> <tr> <td>35 - 39</td> <td>1</td> </tr> </tbody> </table>	Age Group in years	Number of People	5 - 9	1	10 - 14	2	15 - 19	4	20 - 24	7	25 - 29	5	30 - 34	3	35 - 39	1
Age Group in years	Number of People																
5 - 9	1																
10 - 14	2																
15 - 19	4																
20 - 24	7																
25 - 29	5																
30 - 34	3																
35 - 39	1																
6	Emiliano has 22 coins, a mixture of dimes and quarters. If the dimes were quarters and the quarters were dimes, Emiliano would have 60 cents more than he has now. How many quarters does he currently have?																
7	The diagram shows a decagon consisting of 13 unit squares. The portion below $\overline{AB}$ consists of a unit square and a triangle with a base of 5. If $\overline{AB}$ bisects the area of the decagon, then as a reduced common fraction, the area of $\triangle BXY = P/Q$. What is $P + Q$?																

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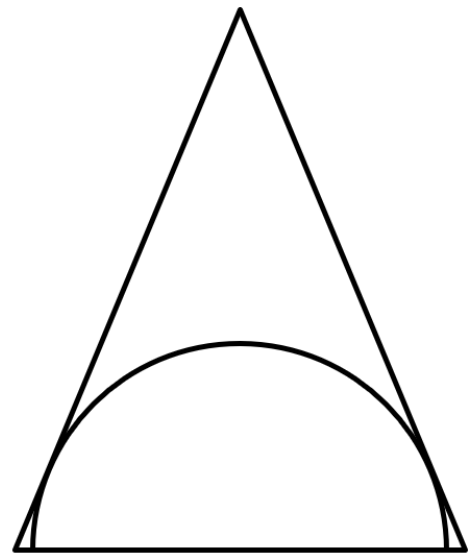
- 8 The numbers 1 through 9 are marked with points on the number line as shown. Two distinct points are chosen at random from these points. As a reduced common fraction, the expected value of the distance between the two points is A/B . What is $A + B$? Note: the expected value is the average of all possible lengths of segments having any two points as endpoints.



- 9 How many triangles of any size are in the figure below?



- 10 A semicircle is inscribed in an isosceles triangle with base 10 cm and height 12 cm so that the diameter of the semicircle is contained in the base of the triangle as shown. As a reduced common fraction, the radius of the semicircle is A/B . What is $A + B$?



"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Key

Linda Moore Triple Jump - Answer Key

7th/8th Grade

Answer	
1	18
2	3
3	100 [plates]
4	[A =] 27
5	24 [years]
6	9 [quarters]
7	[P + Q =] 23
8	[A + B =] 13
9	22 [triangles]
10	[A + B =] 73

1	What is 80% of 22.5?	
2	The integers 1 through 10 are written as English words and listed in order alphabetically. As an integer, what is the ninth word in the list?	
3	It takes 3 workers 20 minutes to wash 80 plates. Working at the same rate, how many plates can 5 workers wash in 15 minutes?	
4	A cylinder has a radius of 15 millimeters and a height of 0.12 meters. In terms of π and in cubic centimeters , the volume of the prism can be expressed as $A\pi \text{ cm}^3$. What is the value of A?	
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6	Emiliano has 22 coins, a mixture of dimes and quarters. If the dimes were quarters and the quarters were dimes, Emiliano would have 60 cents more than he has now. How many quarters does he currently have?	
7	The diagram shows a decagon consisting of 13 unit squares. The portion below $\overline{AB}$ consists of a unit square and a triangle with a base of 5. If $\overline{AB}$ bisects the area of the decagon, then as a reduced common fraction, the area of $\triangle BXY = P/Q$. What is $P + Q$?	
8	The numbers 1 through 9 are marked with points on the number line as shown. Two distinct points are chosen at random from these points. As a reduced common fraction, the expected value of the distance between the two points is A/B . What is $A + B$? Note: the expected value is the average of all possible lengths of segments having any two points as endpoints.	
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"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Submittal #1 score
(out of 10)

Room #

School Name

Team #

Linda Moore Triple Jump - 15 minutes - ~15% of team score

When you are prompted to begin, tear off the three colored answer sheets and give a copy of the test to each of your team members and begin testing. Record all answers on this colored answer sheet. This Submittal #1 will be collected after 5 minutes.

SUBMITTAL #1

STUDENTS: DO NOT WRITE IN SHADED REGIONS

Answer		Scorer 2	Scorer 1
		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
7 th /8 th Grade		TOTAL:	

"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Submittal #2 score
(out of 10)

Room #

School Name

Team #

Linda Moore Triple Jump - 15 minutes - ~15% of team score

This Submittal #2 will be collected after 10 minutes.

SUBMITTAL #2

STUDENTS: DO NOT WRITE IN SHADED REGIONS

Answer		Scorer 2	Scorer 1
		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
7 th /8 th Grade		TOTAL:	

"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Submittal #3 score
(out of 10)

Room #

School Name

Team #

Linda Moore Triple Jump - 15 minutes - ~15% of team score

This Submittal #3 will be collected after 15 minutes.

SUBMITTAL #3

STUDENTS: DO NOT WRITE IN SHADED REGIONS

Answer		Scorer 2	Scorer 1
		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
7 th /8 th Grade		TOTAL:	

"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Room #

School Name

Team #

Total Score for Each Round

College Bowl #1 (10 Possible)	College Bowl #2 (10 Possible)	College Bowl #3 (10 Possible)

DO NOT USE TALLY MARKS ON THIS SHEET. WRITE THE TOTAL SCORE FOR EACH ROUND.

"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Room #

School Name

Team #

Total Score for Each Round

College Bowl #1 (10 Possible)	College Bowl #2 (10 Possible)	College Bowl #3 (10 Possible)

DO NOT USE TALLY MARKS ON THIS SHEET. WRITE THE TOTAL SCORE FOR EACH ROUND.

"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Proctor
Copy

Mental Math Contest

MENTAL MATH - 30 seconds per question - ~25% of team score & ~8% of individual score

Error! Reference source not found.

1	What is the mean of the numbers forty, sixty, eighty, and twenty?	
2	How many millimeters are in zero-point-three-five meters?	
3	A bag contains only pink and purple candies. All but twelve of the candies are pink and all but nine of the candies are purple. How many candies are in the bag?	
4	In inches, what is the perimeter of a rectangle whose length is ten-point-five inches and whose width is six-point-five inches?	
5	How many ways are there to arrange the letters in the word SASS, spelled S-A-S-S?	
6	Anderson completes a one-mile race in six minutes. In this race, Brody runs at an average rate that is three-fourths as fast as Anderson. In seconds, how long does it take Brody to complete the one-mile race?	
7	The rule in a certain number sequence is: divide the first term by one to get the second term, divide the second term by two to get the third term, divide the third term by three to get the fourth term, and so on. If the first term in this sequence is one hundred twenty, what is the sixth term?	
8	What is the least common multiple of ten, twenty-four, and fifty-four?	

"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Key

COLLEGE BOWL ROUND #1

#	Problem	Answer
1	What is the sum of the next two numbers in the geometric sequence that begins with three, six, twelve, and twenty-four?	144
2	Solve for X: fifteen X plus sixty equals two hundred and forty	$[x =] 12$
3	What is two times thirty plus three times four hundred plus four times five thousand?	21260
4	Zeus has ten cards, and each card contains a different one of the first ten prime numbers. If one card is randomly selected, what is the probability in percent that it contains the digit three?	30 [%]
5	Shanteel rides her bike for ten minutes at an average rate of six miles per hour and then for five minutes at an average rate of nine miles per hour. In miles per hour, what is her average rate for the whole fifteen minutes?	7 [mph]
6	The ratio of the perimeters of two similar rectangles is two to five. If the area of the larger rectangle is one hundred twenty-five square inches, what is the number of square inches in the area of the smaller rectangle?	20 [in^2]
7	In how many ways can the letters of the word MATH, spelled M - A - T - H, be arranged such that the first letter is not an M?	18 [ways]
8	What is the sum of the even integers between twenty-five and thirty-one?	84 [= sum]
9	As a simplified fraction, the ratio of four hundred and twenty to fifty-six equals A over B. What is A plus B?	$[A + B =] 17$
10	What is the median of the following numbers? One-half, one-fourth, seven-halves, and eight.	2 [= median]

"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Key

COLLEGE BOWL ROUND #2

#	Problem	Answer
1	In a bag of thirty-six candies, one-third of the candies are red, one-fourth of the candies are green, and one sixth of the candies are blue. How many candies are not red, green, or blue?	9 [candies]
2	What is nine squared minus six squared plus three squared?	54
3	Diego takes eighty minutes to walk four miles. Walking at this same rate, how many minutes would it take him to walk two point five miles?	50 [minutes]
4	A field contains only pigs and chickens. In total there are eighty-six legs and sixty eyes among the animals in the field. How many chickens are in the field?	17 [chickens]
5	Two standard six-sided dice are rolled and the numbers showing are added. As a simplified fraction, the probability that the sum of the two numbers is divisible by six is A over B . What is A plus B ?	$[A + B =] 7$
6	In a certain sequence, each term is equal to two minus the quantity three times the previous term. If the first term is five, what is the third term in this sequence?	41 [= 3 rd term]
7	When the number seventy-seven thousand seven hundred seventy-seven is written out in words, how many times does the letter 'e' appear?	11 [times] [e's]
8	One foot is equivalent to twelve inches. Let two point four inches be called a "TOE". How many TOES are in two feet?	10 [toes]
9	The circumference of a circle with a radius of thirteen point five centimeters is $A\pi$ centimeters, where A is a two-digit integer. What is A ?	27 [= A]
10	What is the smallest product of two distinct, positive three-digit composite numbers?	10200

"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Key

COLLEGE BOWL ROUND #3

#	Problem	Answer
1	What is the product of seventy-six times eighty-four?	6384
2	If January 1, 2026, was a Thursday, how many total Fridays will there be in 2026?	52 [Fridays]
3	What is the mean of the numbers forty, seventy, and eighty-two?	64 [= mean]
4	Jenna runs ninety meters in eighteen seconds. In seconds, how long does it take Kira to run ninety meters at an average rate that is twenty percent faster than Jenna's average rate?	15 [seconds]
5	Solve for X: X over six minus three over five equals two X over fifteen plus seven over ten	[$x =$] 39
6	The least common multiple of two positive integers, A and B, is seventy. What is the smallest possible value of A plus B?	[$A + B =$] 17
7	Five fair coins are tossed and each lands showing either heads or tails. As a simplified fraction, the probability that exactly four of the five coins show the same result is A over B. What is A plus B?	[$A + B =$] 21
8	The perimeter of a regular heptagon is one hundred and twenty-six centimeters. In centimeters, what is the length of one side of the heptagon?	18 [cm]
9	Two distinct positive integers, A and B, are both less than fifty and greater than thirty-six. What is the greatest possible value of A minus B?	[$A - B =$] 12
10	The first three terms of an arithmetic sequence are fifty, one hundred twenty, and one hundred ninety. What is the tenth term in the sequence?	680 [= 10 th term]

"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Key

COLLEGE BOWL ROUND #4

#	Problem	Answer
1	What is the largest three-digit integer in which all three digits are prime numbers?	777
2	How many square inches are in a square yard?	1296 [in ²]
3	When a positive integer N is divided by ten the remainder is eight. When N is divided by nine, the remainder is two. What is the smallest value of N?	38 [= N] [= smallest value of N]
4	A sequence is defined by adding five to the numerator and four to the denominator of a given fraction to get the next term. The last term in the sequence is the first integer term in the sequence. If the first term is one-seventh, how many terms are in the sequence?	7 [terms]
5	A Roomba vacuum cleaner vacuums one room in three-tenths of an hour. At this rate, how many rooms will it vacuum in six hours?	20 [rooms]
6	A fair coin is flipped four times. As a simplified fraction, the probability that no two consecutive flips have the same result is A over B. What is A plus B?	[A + B =] 9
7	Naomi has a weekly set of drills she uses to improve her tennis skills. After finishing three drills she has completed twelve percent of her weekly set. How many drills does she have left to complete for the week?	22 [drills]
8	How many distinct ways can four beads be arranged in a straight line if two of them are identical yellow beads, one is green, and one is blue?	12 [ways]
9	The measures of angles A and B in triangle ABC are thirty-two and seventy-seven degrees. What is the number of degrees in the measure of angle C?	71 [°]
10	Solve for X: seven X plus eleven minus four X equals two to the fifth	[x =] 7

"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Key

COLLEGE BOWL ROUND #5

#	Problem	Answer
1	What is the product of fifty-five times fifty-four?	2970
2	A certain animal shelter houses only cats and dogs. If there are a total of one hundred and twelve animals at the shelter and four-sevenths of them are dogs, how many cats are in the shelter?	48 [cats]
3	The mean of three numbers is eighteen. When a fourth number is added to the set, the mean becomes nineteen. What is the fourth number?	22 [= 4 th number]
4	What is two thousand eight hundred and twenty-nine divided by one hundred and twenty-three?	23
5	As a percent, what is the common difference in the arithmetic sequence whose first three terms are one over twenty-five, three over twenty-five, and one-fifth?	8 [%]
6	The letters in the word COTTON, spelled C-O-T-T-O-N, are used to form a six-letter arrangement. As a simplified fraction, the probability that the letters are in alphabetical order is A over B. What is A plus B?	[A + B =] 181
7	Safran can paint ten boards in thirty minutes. Cayenne can paint eight boards in twenty minutes. In minutes, how long would it take the two working together to paint thirty-three boards?	45 [minutes]
8	The numbers from one to twenty-one are written on a piece of paper. How many times does the digit two appear?	4 [times]
9	Solve for X: two X minus seventy-two equals the square root of four hundred.	[x =] 46
10	A rectangular prism has a length of ten centimeters, a width of seven centimeters, and a height of thirteen point five centimeters. In cubic centimeters, what is the volume of the prism?	945 [cm ³]

"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Key

COLLEGE BOWL ROUND #6

#	Problem	Answer
1	Beyoncé reads twelve pages in nine minutes. At this rate, how many pages will she read in twenty-four minutes?	32 [pages]
2	What is the greatest common factor of one hundred twenty-six and two hundred ten?	42 [= gcf]
3	Let A , B , and C represent three distinct positive integers, all less than sixteen. If A is greater than B and C is less than B , what is the greatest value of B minus C ?	$[B - C =] 13$
4	A number is divided by negative six, then the quotient is increased by six. The result is two. What was the original number?	24 [= number]
5	A rectangle has an area of three square feet, and its length is twenty-seven times its width. What is the number of inches in the perimeter of the rectangle?	224 [inches]
6	Let A be fifteen percent of eighty and let B be twenty-five percent of one hundred and twenty. As a percent, what is A divided by B ?	40 [%]
7	A data set consists of six positive integers and has a unique mode. If the mean is twelve, what is the greatest possible value of the mode?	31 [= mode]
8	Two plinks are equal to six winks. Three winks are equal to ten zinks. How many zinks are equal to five plinks?	50 [zinks]
9	What is the sum of the integers three through twenty-three?	273 [= sum]
10	There are one hundred centimeters in a meter. How many cubic centimeters are equivalent to three-thousandths of a cubic meter?	3000 [cm^3]

"Math Is Cool" Championships — 2025-26

7th/8th Grade — January/February 2026

Key

COLLEGE BOWL EXTRA

#	Problem	Answer
1	What is the sum of the first six positive perfect cubes?	441 [= sum]
2	How many quarters does it take to make seventeen dollars and fifty cents?	70 [quarters]
3	What is the sixth term in the geometric sequence that begins with the three terms ninety-six, forty-eight, and twenty-four?	3 [= 6 th term]
4	How many meters are in one point seven five kilometers?	1750 [meters]
5	A regular hexagon with a perimeter of sixty-six centimeters has a diagonal drawn parallel to two of its sides. In centimeters, how long is this diagonal?	22 [cm]
6	What is the median of the numbers six, three, fifteen, eleven, seven, and nine?	8 [= median]
7	A car drives six hundred thirty miles on thirty-five gallons of gas. How many miles can it drive on twelve gallons of gas?	216 [miles]
8	There are fifty unpaired socks in a sock drawer. Fifteen are pink, fifteen are purple, and the rest are yellow. How many socks do you have to take out of the drawer to have guaranteed that at least one sock of each color was taken out?	36 [socks]
9	What is one thousand two hundred thirty-seven point seven eight five rounded to the nearest hundred?	1200 [12 hundred]
10	A square piece of paper is folded in half horizontally to form a rectangle. The rectangle has a perimeter of eighteen inches. In inches, what is the perimeter of the original square?	24 [inches]